

		ball's velocity
2	C	Height of X above horizontal surface = area under $v - t$ graph from $t = 0$ to $t = 1.0$ s $= \frac{1}{2} \times 1.0 \times 10.0$ $= 5.0 \text{ m}$ Height reached by ball after first rebound = area under $v - t$ graph from $t = 1.0$ s to $t = 1.5$ s $= \frac{1}{2} \times 0.50 \times 5.0$ $= 1.25 \text{ m}$ Ratio $\frac{\text{magnitude of displacement of the ball from X at 1.5 s}}{\text{height of X above horizontal surface}}$ $= \frac{5.0 - 1.25}{5.0}$ $= 0.75$
3	D	Using relative speed of approach = relative speed of separation